

PROJECT 1: Dockerizing Jenkins Pipeline

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Figure 1: Commit of Project 1 Jenkins Pipeline files to the Git repository

```
AnushkaDesai> git show --stat --oneline HEAD
31bc721 (HEAD -> Devops_LabL1_2023-27, origin/HEAD, origin/Devops_LabL1_2023-27) Added Project 1 Jenkins Pipeline
.../README.md | 0
.../my-flask-app/Dockerfile | 11 ++++++++
.../my-flask-app/Jenkinsfile | 25 ++++++++
+++++++
.../my-flask-app/app.py | 11 ++++++++
.../my-flask-app/requirements.txt | 1 +
5 files changed, 48 insertions(+)
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_
```

Figure 2: Successfully pushing Project 1 Jenkins Pipeline to GitHub

```
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_
AnushkaDesai> git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 1.12 KiB | 127.00 KiB/s, done.
Total 10 (delta 2), reused 2 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/AnushkaDesai13/Devops-Lab-L1_2023-27.git
 0afcd4f..31bc721 Devops_LabL1_2023-27 -> Devops_LabL1_2023-27
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_
AnushkaDesai>
```

Figure 3: Project 1 source code and Jenkinsfile uploaded to GitHub

← Files

Devops_LabL1_20...

...

[Devops-Lab-L1_2023-27](#) / [23070122035_AnushkaDesai](#)
/ [Project1 Dockerizing Jenkins Pipeline](#)
/ [my-flask-app](#) /

 **AnushkaDesai13** Added Project 1 Jenkins Pipeline

31bc721 · last week

This branch is [5 commits ahead of](#) and [209 commits behind](#)
[aditisharmas11/Devops-Lab-L1_2023-27:Devops_LabL1_2023-27](#) .

 **Contribute**

 **Sync fork**

Name	Last commit message	Last commit date
 ..		
 Dockerfile	Added Project 1 Jenkins Pipeline	last week
 Jenkinsfile	Added Project 1 Jenkins Pipeline	last week
 app.py	Added Project 1 Jenkins Pipeline	last week
 requirements.txt	Added Project 1 Jenkins Pipeline	last week

Figure 4: Jenkins LTS Container Running Successfully Using Docker

```
hell
Microsoft Corporation. All rights reserved.

system32> Stop-Process -Id 8984 -Force
system32> netstat -ano | findstr :8080
system32> docker ps -a

```

IMAGE	COMMAND	CREATED	STATUS	PORTS
jenkins/jenkins:lts-jdk17	"/usr/bin/tini -- /u..."	7 minutes ago	Created	
myjenkins				
hello-world-flask	"python app.py"	13 days ago	Exited (255) 9 days ago	0.0.0.0:5000->0.0.0.0:5000
flaskapp				
3a0f252af6aa	"python app.py"	13 days ago	Exited (0) 13 days ago	
charming_haibt				

```
system32> docker start myjenkins

system32> docker ps

```

IMAGE	COMMAND	CREATED	NAMES	STATUS	PORTS
jenkins/jenkins:lts-jdk17	"/usr/bin/tini -- /u..."	8 minutes ago	myjenkins	Up 2 seconds	0.0.0.0:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp

```
system32>
```

Figure 5: Creating the First Jenkins Administrator User

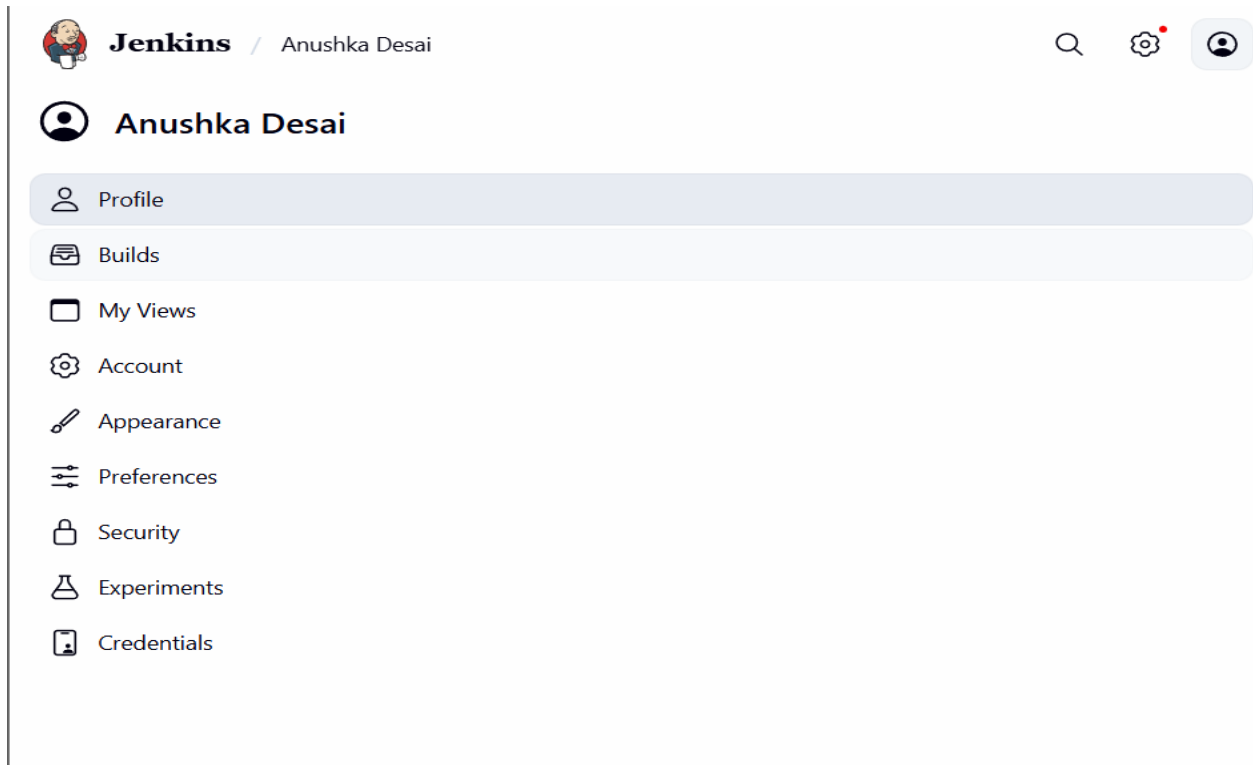


Figure 6: Jenkins Setup Completed and Dashboard Ready

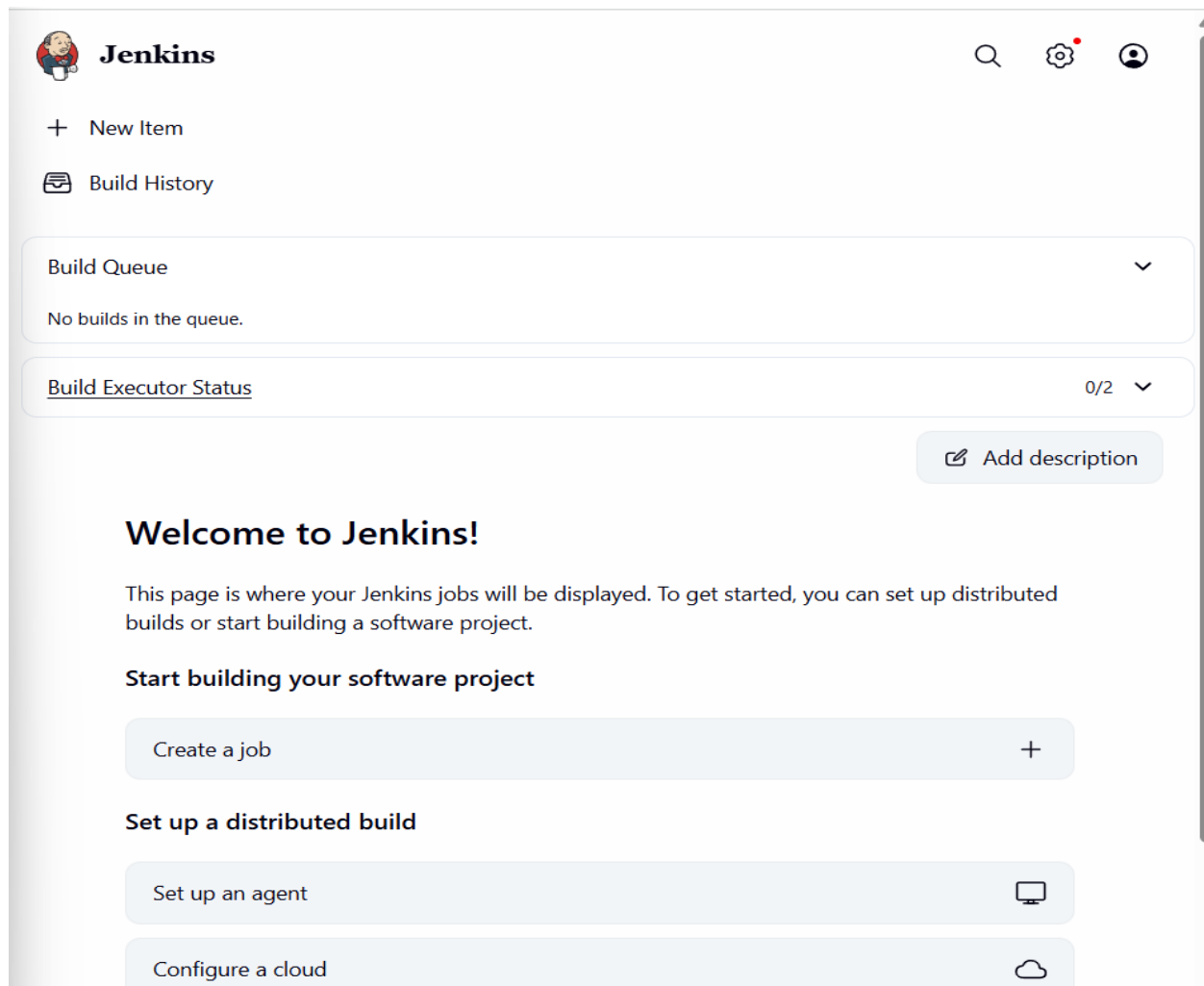


Figure 7: Creating a Jenkins Pipeline Project

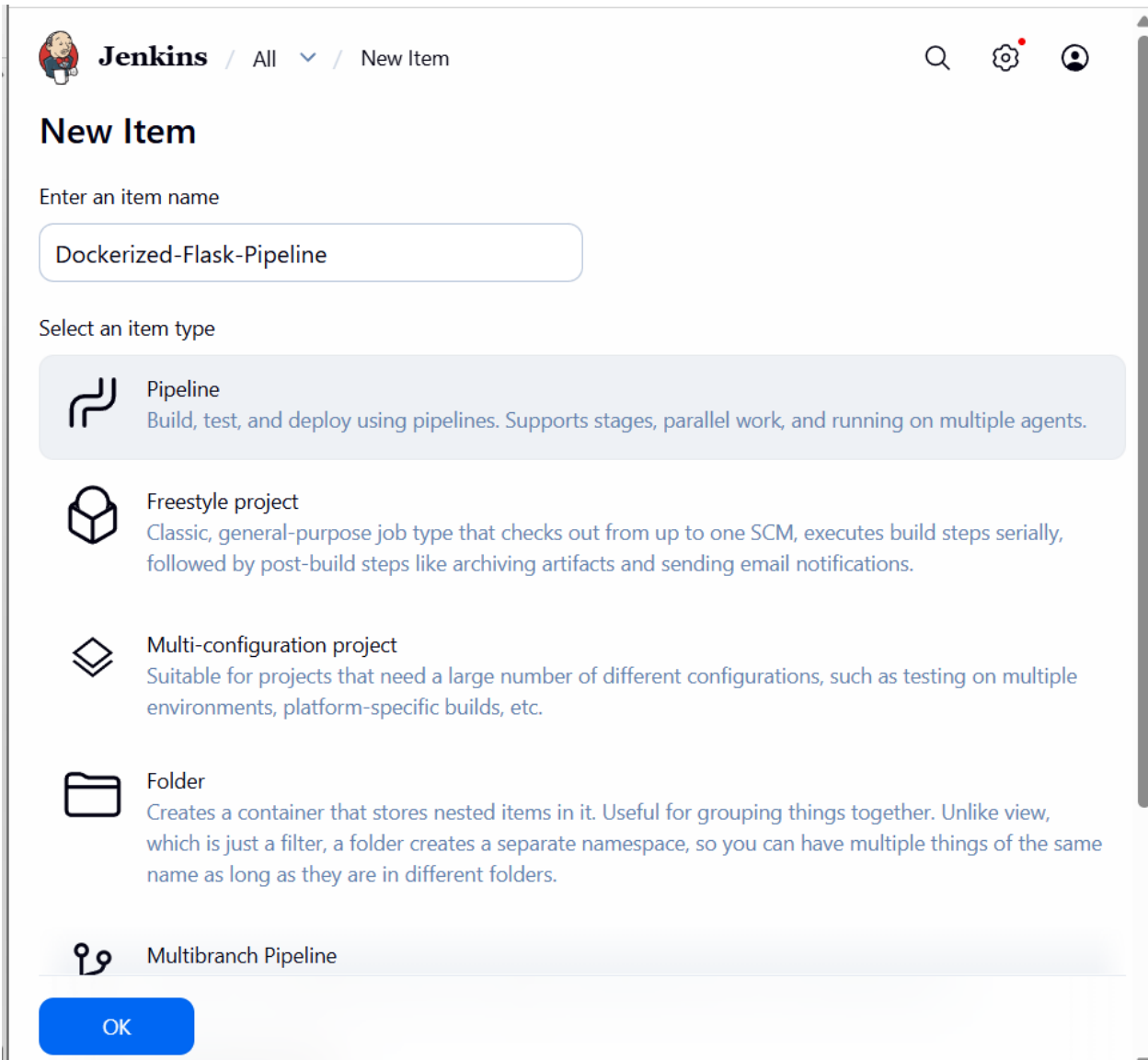


Figure 8: Configuring Jenkins Pipeline to Use the Jenkinsfile from GitHub

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

None

Script Path ?

Jenkinsfile

☒ Lightweight checkout ?

[Pipeline Syntax](#)

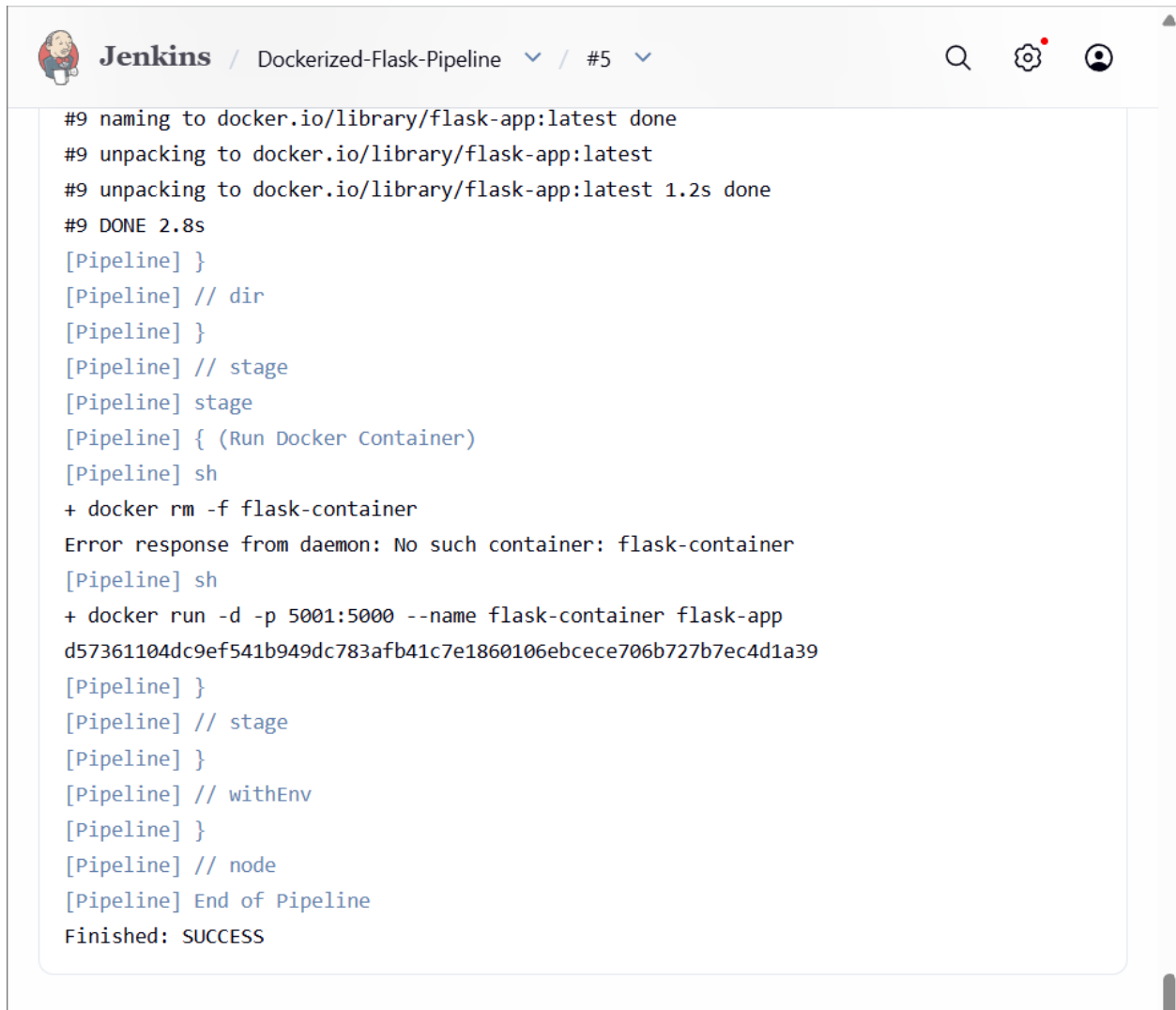
Advanced

Figure 8: Updated Jenkinsfile for Linux-based Jenkins Docker Pipeline

Project1_Dockerizing_Jenkins_Pipeline > my-flask-app >  Jenkinsfile

```
1 pipeline {
2     agent any
3
4     stages {
5
6         stage('Checkout') {
7             steps {
8                 echo 'Checking out source code...'
9             }
10        }
11
12        stage('Build Docker Image') {
13            steps {
14                sh 'docker build -t flask-app .'
15            }
16        }
17
18        stage('Run Docker Container') {
19            steps {
20                sh 'docker rm -f flask-container || true'
21                sh 'docker run -d -p 5001:5000 --name flask-c'
22            }
23        }
24    }
25 }
```

Figure 9: Successful Execution of Dockerized Flask Jenkins Pipeline



```
Jenkins / Dockerized-Flask-Pipeline / #5

#9 naming to docker.io/library/flask-app:latest done
#9 unpacking to docker.io/library/flask-app:latest
#9 unpacking to docker.io/library/flask-app:latest 1.2s done
#9 DONE 2.8s
[Pipeline] }
[Pipeline] // dir
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Run Docker Container)
[Pipeline] sh
+ docker rm -f flask-container
Error response from daemon: No such container: flask-container
[Pipeline] sh
+ docker run -d -p 5001:5000 --name flask-container flask-app
d57361104dc9ef541b949dc783afb41c7e1860106ebcece706b727b7ec4d1a39
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Figure 10: Flask Application Successfully Deployed in Docker Container through Jenkins Pipeline

